NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) Date: 12-04-26
Time: 3 Hrs Max. Marks: 180

12-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-5(P2)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: TOTAL SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

THE NARAYANA GROUP

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TIME: 3HRS **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 6)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 7 – 12)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 13 – 16)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	–1	4	12
Sec – IV (Q.N : 17 – 19)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 20 – 25)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 26 – 31)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 32 – 35)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	–1	4	12
Sec – IV (Q.N : 36 – 38)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 39 – 44)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 45 – 50)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	–1	4	12
Sec – IV (Q.N : 55 – 57)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

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MATHEMATICS **MAX.MARKS: 60**

SECTION - 1 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

1. $\int \frac{x+2}{\sqrt{x^4+8x^3+20x^2+16x+4}}dx = f(x)+C$ and $f(0)=\frac{\ell n2}{2}$ then which is/are in correct?

A) Number of solutions of $f(x)=x^2$ is 2

B) Number of solutions of $f(x)=x^2$ is 1

C) Number of solutions of $f(x)=\sqrt{4-x^2}$ is 2

D) Number of solutions of $f(x)=\sqrt{4-x^2}$ is 1

2. If the minimum of the following function $f(x)$ defined at $0<x<\frac{\pi}{2}$, $f(x)=\int_0^x \frac{d\theta}{\cos\theta}+\int_x^{\frac{\pi}{2}} \frac{d\theta}{\sin\theta}$ is equal to $\ln\left(a+\sqrt{b}\right)$ where $a,b\in N$ and b is not a perfect square then the value of $(a+b)$ is not divisible by

A) 2 B) 3 C) 5 D) 7

3. Let a,b,c be distinct non-zero complex numbers with $|a|=|b|=|c|$. If each of the equations $az^2+bz+c=0$ and $bz^2+cz+a=0$ has root having modulus 1 then which is/are correct

A) $b^2=ac, c^2=ab$ B) $a^2+b^2+c^2=0$

C) $(a-b)^2+(b-c)^2+(c-a)^2=0$ D) $|a-b|=|b-c|=|c-a|$

4. Let $A(0,0), B(b,0), C(0,c)$ be vertices of a triangle. If $BC=a$ and BC is divided into n equal parts (segments) (n odd integer). Let 'θ' be the angle subtended by that segment of hypotenuse BC, which contains the mid point of BC at A. If h be the length of the altitude to the hypotenuse of the triangle, and if $\tan\theta=\frac{pnh}{(qn^2-r)a}$ where $p,q,r\in N$ and $G.C.D(p,q,r)=1$ then which is/are correct

A) $p>q$ B) $p<q$ C) $p+q=5$ D) $r=2$

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5. $L_1: y-2=z+3=0,\ L_2: z-3=x+1=0,\ L_3: x-1=y+2=0$, be 3 lines in three dimensional geometry. Then which is/are incorrect

A) L_1, L_2, L_3 cannot be coplanar

B) L_1, L_2, L_3 can form an equilateral triangle

C) Two of them are perpendicular

D) L_1, L_2, L_3 are skew lines pairwise

6. The probability that a 7-digit number whose sum of digits is 59 is divisible by 11 is $\frac{a}{b}$ and $\left(GCD(a,b)=1\right)$ then the value of $a+b$ is k, then which is/are correct

A) k is square of a integer

B) k is a two digited number

C) k is a multiple of 10

D) k is a composite number

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 7 and 8:

Consider the equation $(\cos x-\sin x)\left(2\tan x+\frac{1}{\cos x}\right)+2=0$

7. Number of solutions in $(0,4\pi)$ is

8. Number of solutions in $(0,40\pi)$ is

Question Stem for Question Nos. 9 and 10:

Let $f: R\to R;\ g: R\to R$ be the differentiable functions defined as

$f(x)=x^2+xg'(1)+g''(2),\ g(x)=x^2f(1)+xf'(x)+f''(x)$

9. The value of $f(10)$ is equal to _____

10. The value of $|g(6)|$ is equal to _____

Question Stem for Question Nos. 11 and 12:

Let A and B are square matrices of order 2 with real elements. Such that

$AB=A^2B^2-(AB)^2$ and $\det(B)=3$

11. The value of $\det(A+2B)-\det(B+2A)$ is equal to _____

12. The value of $\det(A-4B)-\det(B-4A)$ is equal to _____

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SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer. For each question, choose the option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.
Negative Marks: –1 In all other cases.

Question Stem for Question Nos. 13 and 14:

System of equations can be evaluated by many methods. Some of them are

1) Matrix Inversion Method

2) Cramer's Method

3) Gauss – Jordon Method

But Matrix Inversion Method is not helpful always because it cannot be used for the case of no solution and infinitely many solutions.

The remaining 2 methods are helpful to evaluate all the possibilities that the system of equations will have

13. Let α,β,γ be real roots of the equation,

$x^3+ax^2+bx+c=0, (a,b,c\in R \text{ and } a,b\neq 0)$. If the system of equations (in u,v,w) given by $\alpha u+\beta v+\gamma w=0;\ \beta u+\gamma v+\alpha w=0;\ \gamma u+\alpha v+\beta w=0$ has non-trivial solution, then the minimum value of $x(3x+2a)+b+1$

A) 5 B) 3 C) 1 D) 0

14. Consider the following system of equations

$x+2y-3z=a;\ 2x+6y-11z=b;\ x-2y+7z=c$,

Where a,b and c are real constants. Then the system of equations

A) has a unique solution when $5a=2b+c$

B) has infinite number of solutions when $5a=2b+c$

C) has no solution for all a,b and c

D) has a unique solution for all a,b and c

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Question Stem for Question Nos. 15 and 16:

Let $X_1, X_2, ..., X_{18}$ be eighteen observations such that $\sum_{i=1}^{18}(X_i - \alpha) = 36$ and $\sum_{i=1}^{18}(X_i - \beta)^2 = 90$, where α and β are distinct real numbers. If the standard deviation of these observations is 1,

15. The value of $|\alpha - \beta|$ is _____

A) 2 B) 4 C) 5 D) 8

16. If $A = \{x \in R : |x - |\alpha - \beta| + 2| > 1\}$, $B = \{x \in R : |\sqrt{x^2 - |\alpha - \beta| + 1}| > 1\}$, $C = \{x \in R : |x - |\alpha - \beta|| \geq 2\}$ and Z is the set of all integers, then the number of subsets of the set $(A \cap B \cap C)^C \cap Z$ is _____ (P^C is compliment of set P)

A) 32 B) 64 C) 128 D) 256

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

17. Let $S = \{1,2,3,4,.....,10\}$, Let $A, B \subseteq S$. Then the number of ordered pairs (A,B) satisfying $(i)\ n(A) \leq n(B)$ $(ii)\ n(A \cap B) = 4$ $(iii)\ A \cup B = S$ is equal to k then maximum digit in k is

18. Let $z = 10 + (n)^{\frac{1}{4}} i$ be a complex numbers, $n \in N$ then for the number $|z| + \sqrt{(\text{Re}(z) + \text{Im}(z))(\text{Re}(z) - \text{Im}(z))}$ to be integer the smallest positive integral value of n equal to k then smallest digit in k is equal to_____

19. The minimum value of $\left[(t-r)^2 + \left(12 - \sqrt{1-t^2} - \sqrt{4r}\right)^2\right]^{\frac{1}{2}}$ for all permissible values of t and r is equal to $a\sqrt{b} - c$ where $a,b,c \in N$ (b is a prime number) then the value of $a + b - c - 4$ is equal to

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PHYSICS **MAX.MARKS: 60**

SECTION - 1 (Maximum Marks : 24)

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Negative Marks: -2 In all other cases.

20. A variable capacitor with initial capacitance C_0 is charged to a voltage V_0 and is short circuited through a resistance R. If current remains constant during discharging (an external agent is pulling the plates apart) then choose correct option(s)

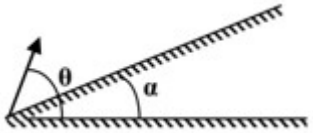
A) Variation of capacitance with time is given by $C = C_0\left(1 - \frac{t}{2RC_0}\right)$

B) Time taken to discharge the capacitor is given by $t = RC_0$

C) Total heat produced in the resistor is $C_0V_0^2$

D) Work done by external agent is $\frac{3}{2}C_0V_0^2$

21. A perfectly elastic particle is projected with velocity v at an elevation θ. A smooth plane passes through the point of projection and is inclined at an angle α to the horizontal. The particle will return to the point of projection provided $\cot\alpha\cot(\theta-\alpha)$ is:



A) $3\sqrt{3}$ B) 4.5 C) 6 D) 3

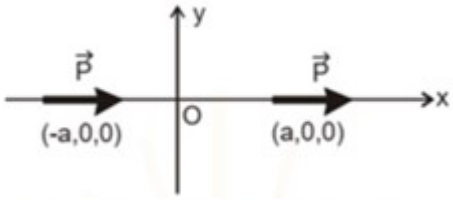
22. A thin uniform square plate of side $60\ cm$ suspended by pivoting at a point on its periphery and made to oscillate in the vertical plane coinciding the plane of square plate. The possible values of time period of small angular oscillations of the plate is / are_____

[take $g = \pi^2\frac{m}{s^2}$]

A) 1.4 sec B) 1.5 sec C) 1.6 sec D) 1.7 sec

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23. Two identical dipoles of dipole moment $\vec{P} = P_0 i$ (P_0 is a positive constant) are placed on $x-$axis at points $A(a,0,0)$ and $B(-a,0,0)$ as shown. Then pick up the correct statements:



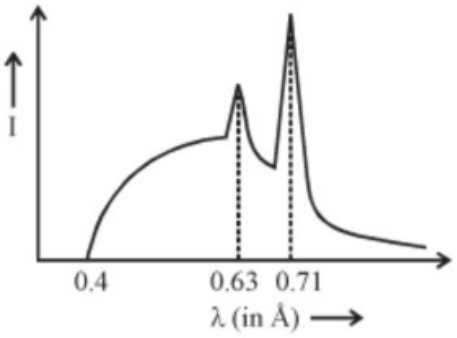
A) The electric field at each point on $y-z$ plane (except at infinity) must be perpendicular to $y-z$ plane.

B) If electric field exists at a point on $y-z$ plane, it must be perpendicular to $y-z$ plane.

C) Potential at each point on $y-z$ plane is zero.

D) There is a circle of finite radius on $y-z$ plane with centre at origin such that both electric field and potential are zero at each point on its periphery.

24. Figure shows intensity versus wavelength graph of X-rays coming from Coolidge-tube with molebdenum as anticathode. The two peaks shown in graph corresponds to K_α & K_β X-rays



A) Wavelength of L_α X-rays from Coolidge tube will be (approximately) $5.60\,\text{Å}$

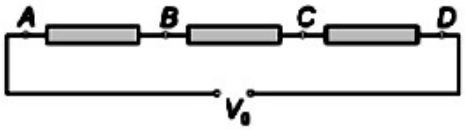
B) Voltage applied across Coolidge tube is (approximately) $16\ kV$

C) If the potential difference across the Coolidge tube is increased such that the cut off wavelength of X-ray becomes $0.2\,\text{Å}$. Then the value of $(\lambda_{k\alpha} - \lambda_{k\beta})$ in this condition is $0.08\,\text{Å}$

D) If cut off wavelength is increased to $0.65\,\text{Å}$, intensity of K_α & K_β must increase

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25. Three resistors are connected in series with a battery of internal voltage $V_0 = 62V$ and internal resistance $R_b \approx 0$ as shown in the figure, then measurements are carried out with a single voltmeter. The results of the first three measurements are: $V_{AB} = V_{BC} = V_{CD} = 20V$. What does the instrument show when it is connected between point pairs AC and AD?



A) $V_{AC} = V_{BD} = 40V$ B) $V_{AC} = V_{BD} \neq 40V$ C) $V_{AD} = 60V$ D) $V_{AD} \neq 60V$

SECTION - 2 (Maximum Marks : 12)

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Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 26 and 27:

A sprinkler in a lawn is rotating at a constant angular speed $\omega = 0.5\ rad/s$. Water comes out of the hole at a constant speed $u = 20\,m/s$ and at angle 45^0 above horizontal. Neglect the height from ground level and arm length of the sprinkler. $\left(g = 10 m/s^2\right)$

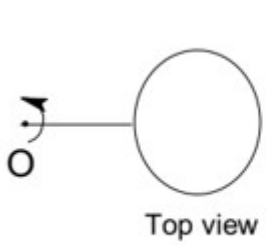


26. The radius of curvature of the trajectory of a water particle at its highest point is__ metre

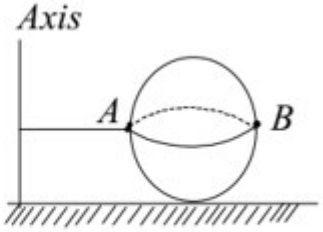
27. The radius of curvature of the water trail at its highest point is ____ metre

Question Stem for Question Nos. 28 and 29:

A uniform smooth sphere mass 'm' Radius 'R' is being rotated uniformly with constant angular speed 'ω' on a smooth horizontal table about a vertical axis passing through end 'O' with the help of a rigid frame as shown in figure. Rigid frame is made up of a thin rod mass 'm' length 'R' and a ring mass 'm' Radius 'R'. Centre of sphere lies in the plane of frame



Top view



Front view

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28. Angular momentum of system about axis of rotation is _____ $kg\ m^2$ / sec

$\left(m=\frac{3}{7}\ kg,\ R=1m,\ \omega=1\ rad/\sec\right)$

29. There is a point 'A' at the end of rod connected with ring. There is another point 'B' which lies on the surface of the sphere in plane of frame. At the instant shown in figure points 'A' and 'B' are at maximum possible distance. Instantaneous acceleration of 'A' with respect to 'B' will be _____

$\left(m=\frac{3}{7}\ kg, R=1m,\ \omega=1 rad/\sec\right)$

Question Stem for Question Nos. 30 and 31:

A solid metallic sphere of radius r is enclosed in a thin metallic spherical shell of radius $2r$. The spherical shell is kept inside a chamber whose outer walls are completely reflective and adiabatic. Emissivity of the spherical shell is same everywhere and is equal to 0.6. Emissivity of the solid sphere is 0.8. Heat transfer can take place only through radiation. Sphere and shell are in thermal equilibrium with each other and the chamber and the temperature of the shell is T. Assume both spheres are opaque.

30. Energy absorbed by the shell per unit time is $n\sigma\pi r^2T^4$, then $n=$

31. Energy reflected on the solid sphere per unit time is $n\sigma\pi r^2T^4$, then $n=$

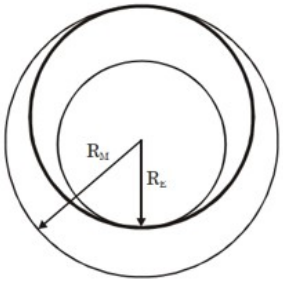
SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer. For each question, choose the option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.
Negative Marks: −1 In all other cases.

Question Stem for Question Nos. 32 and 33:

The space ship starts out in a circular orbit around the sun very near the Earth and has a goal of moving to a circular orbit around the Sun that is very close to Mars. It will make this transfer in an elliptical orbit as shown in bold in the diagram below. This is accomplished with an initial velocity boost near the Earth Δv_1 and then a second velocity boost near Mars Δv_2. Assume that both of these boosts are from instantaneous impulses, and ignore mass changes in the rocket as well as gravitational attraction to either Earth or Mars. Don't ignore the Sun! Assume that the Earth and Mars are both in circular orbits around the Sun of radii R_E and $R_M = R_E/\alpha$ respectively. The orbital speeds are V_E and V_M respectively.



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32. Derive an expression for the velocity boost Δv_1 to change the orbit from circular to elliptical. Express your answer in terms of V_E and α:

A) $\Delta v_1 = v_E\left(\sqrt{\frac{2}{1+\alpha}}+1\right)$ B) $\Delta v_1 = v_E\left(2\sqrt{\frac{1}{1+\alpha}}-1\right)$

C) $\Delta v_1 = v_E\left(\sqrt{\frac{2}{1+\alpha}}-1\right)$ D) $\Delta v_1 = v_E\left(\sqrt{\frac{1}{2(1+\alpha)}}-1\right)$

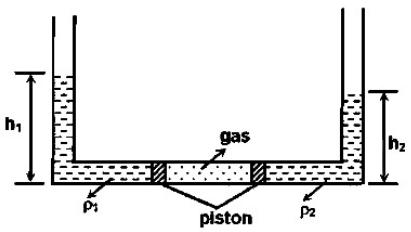
33. Derive an expression for the velocity boost Δv_2 to change the orbit from elliptical to circular. Express your answer in terms of v_E and α:

A) $\Delta v_2 = v_E\sqrt{\alpha}\left(1+\sqrt{\frac{2}{1+\frac{1}{\alpha}}}\right)$ B) $\Delta v_2 = v_E\sqrt{\alpha}\left(1-\sqrt{\frac{1}{2\left(1+\frac{1}{\alpha}\right)}}\right)$

C) $\Delta v_2 = v_E\sqrt{\alpha}\left(1-\sqrt{\frac{1}{1-\frac{1}{\alpha}}}\right)$ D) $\Delta v_2 = v_E\sqrt{\alpha}\left(1-\sqrt{\frac{2}{1+\frac{1}{\alpha}}}\right)$

Question Stem for Question Nos. 34 and 35:

Figure shows a tube whose two limbs are vertical and one is horizontal of sufficient length. A monoatomic ideal gas is enclosed between two massless non conducting movable piston of cross-sectional area 100 cm^2. Liquids of density $\rho_1 = 1000\ kg/m^3$ and $\rho_2 = 2000\ kg/m^3$ are poured in the left and right vertical limbs respectively such that the heights of $h_1 = 4\ m$ and $h_2 = 2\ m$ of the liquids ρ_1 and ρ_2 are formed in equilibrium with separation between the pistons 4 m. Now gas is slowly heated. Consider the wall of tube is insulated and cross-sectional area of the tube is same everywhere and is equal to 100 cm^2 $P_0 = 10^5\ Pa$. If due to slow heating, separation between the piston is slowly increased by 3 m, then



34. Work done by the gas is

A) 3600 J B) 4500 J C) 9000 J D) 7200 J

35. Heat supplied to the gas is

A) 9700 J B) 14400 J C) 12900 J D) 8500 J

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SECTION - 4 (Maximum Marks : 12)

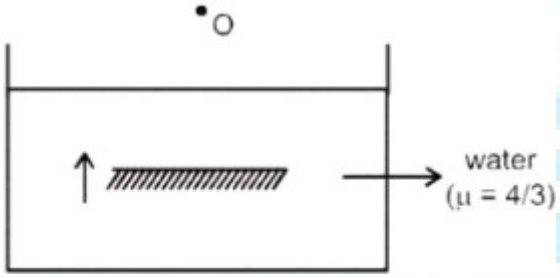
This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
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Zero Marks: 0 In all other cases.

36. Suppose that one of the slits in YDSE is wider than the other, so the amplitude of the light reaching on a screen from, one slit, acting alone is thrice that from the other slit, acting alone. The intensity for Central Bright Fringe is $I_0 = 25\ W/m^2$. Then find the intensity $I\left(W/m^2\right)$ on the screen for (angular position) $\theta = 37^0$. (Take $\frac{d}{\lambda} = \frac{5}{6}$, where d is the separation between the slits and λ is the wavelength of the light)

(Assume the wavefronts coming out of slits to be nearly spherical and screen is thin and far away).

37. A conducting ring of mass $4\ kg$, radius $1m$ and resistance $8\ \Omega$ is placed on a rough horizontal plane xy. There exists a uniform magnetic field $\overline{B} = 4\hat{i} + 4t^2\hat{k}$ Tesla. At what time (in sec) the ring will start toppling? (assume that the ring does not slip)
$\left(g = \pi^2 m/s^2\right)$

38. Mirror in the arrangement shown in figure is moving up with speed $4\ cm/\text{sec}$. Find the speed of final image of object O $(in\ cm/s)$ formed after two refraction and one reflection. $(\mu = 4/3)$



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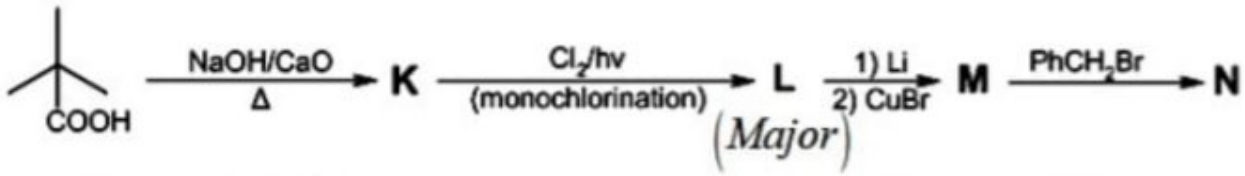
CHEMISTRY **MAX.MARKS: 60**

SECTION - 1 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

39. Consider the following reaction sequence:



Choose the correct option

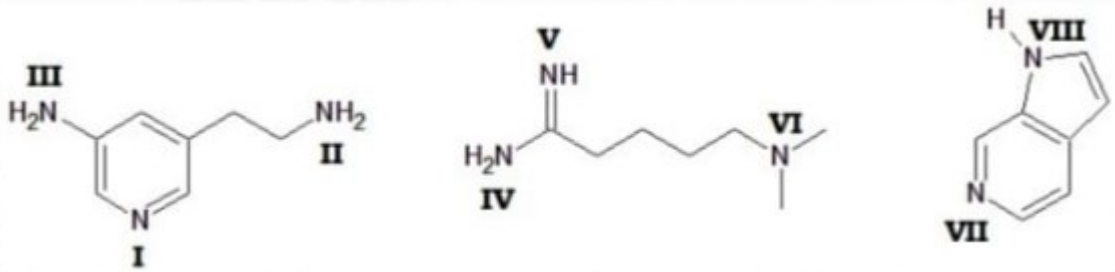
A) N has 9 primary hydrogen atoms

B) K reacts with $KMnO_4$ to give isobutyric acid

C) N on reaction with Br_2 / light gives 7 isomeric products of which only one is achiral

D) All hydrogen atoms in M are primary

40. Consider the following compounds and the nitrogen atoms labelled from I to VIII



Choose the correct statement(s) from the following:

A) If an equimolar mixture of all these three compounds is treated with HCl, V gets protonated first

B) Basic character of I, II and III follow the order II > I > III

C) The lower basic nature of VIII compared to VII can be attributed to involvement of lone pair on VIII in aromatic stabilization

D) Basic character of I, V and VII will almost be equal as all them are sp^2 hybridized

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41. In the following gaseous phase first order reaction

$$A(g) \rightarrow 2B(g) + C(g)$$

Initial pressure was found to be 400 mm of Hg and it changed to 1000 mm of Hg after 20 min. Then which of the following is/are correct? ($\ln 2 = 0.693$)

A) Half life for A is 10 min

B) Rate constant is $0.0693\,min^{-1}$

C) partial pressure of C at 30 min is 350 mm of Hg

D) total pressure after 30 min is 1100 mm of Hg

42. Which of the following statement(s) is/are **CORRECT**?

A) $CH_4 < NF_3 < NH_3 < H_2O$ (Dipole moment order)

B) Order of bond angle: $Cl_2O > F_2O$

C) Trisilyl amine > Trimethyl amine (Lewis basic nature)

D) $NO_2^+ > NO_3^- > NO_2^-$ (Bond angle)

43. Choose the **incorrect** statements

A) Correct order of Δ_O value: $\left[Co(NH_3)_6\right]^{3+} > \left[Rh(NH_3)_6\right]^{3+} > \left[Ir(NH_3)_6\right]^{3+}$

B) Order of spin – only magnetic moment value:

$$\left[Co(H_2O)_6\right]^{2+} < \left[Cr(H_2O)_6\right]^{3+} < \left[Fe(H_2O)_6\right]^{2+}$$

C) Among $\left[V(CO)_6\right]^-$, $\left[Mn(CO)_6\right]^+$, $\left[Fe(CO)_5\right]$ and $\left[Ni(CO)_4\right]$; 'carbon – oxygen' bond length is maximum in $\left[Mn(CO)_6\right]^+$

D) $\left[Co(CN)_6\right]^{3-} > \left[Co(NH_3)_6\right]^{3+} > \left[Co(H_2O)_6\right]^{3+}$: correct order of absorbed wave length of light in the visible region

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44. $MnO_2 + 4HCl(conc.) \xrightarrow{\Delta} A + B(g) + 2H_2O$

$4FeS_2 + 11O_2 \xrightarrow{\Delta} C(s) + D(g)$

$B(g) + D(g) \xrightarrow{charcoal} E(l)$

$P_4 + E \rightarrow F + D(g)$

Which of the following are correct about the compounds of the above reactions.

A) 'B'(g) can act as bleaching agent which bleaches due to reduction in the presence of moisture.

B) The compound 'D' can act as both Lewis acid and Lewis base.

C) $D(g)$ can act as both oxidant and reductant.

D) The compound 'F' on complete hydrolysis give a tri basic acid and a mono basic acid.

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 45 and 46:

An aromatic organic compound A (Molecular formula C_8H_{10}) gives 3 different products on reductive ozonolysis when compound A react with $H^{\oplus} / KMnO_4$ gives B, which on heating with NH_3 gives C. On further strong heating of C gives compound D. Compound D react with $NaOH + Br_2, \Delta$ followed by acidification gives compound E, when compound D react with (i) KOH (ii) $Et - Br$ followed by acidic hydrolysis gives compound B and F. Reduction of compound D by $LiAlH_4$ gives compound G. G further react with $NaNO_2 + HCl$ gives a yellow liquid of compound H.

Based on above information answer the following questions.

[Given molecular weight $C = 12, O = 16, N = 14, H = 1, Br = 80, Na = 23, K = 39$]

45. Ratio of double bond equivalent of compound H / compound A is ____

46. Sum of molecular weight of product E and F is _____

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Question Stem for Question Nos. 47 and 48:

One mole of oxygen gas at 300K in thermal contact with surroundings expands isothermally from 1L to 2L at constant external pressure of 1 atm. Choose the correct option(s) for the process:

(R = 8 J/Kmol, 1 Latm = 101.3 J, C_p of O_2 is $\frac{5}{2}R$, $\ln 2 = 0.7$)

47. The value of ΔS_{sys} (in $JK^{-1}mol^{-1}$) is---------

48. The magnitude of the value of ΔG_{sys} (in $kJ.mol^{-1}$) is--------- (rounded to two decimal places)

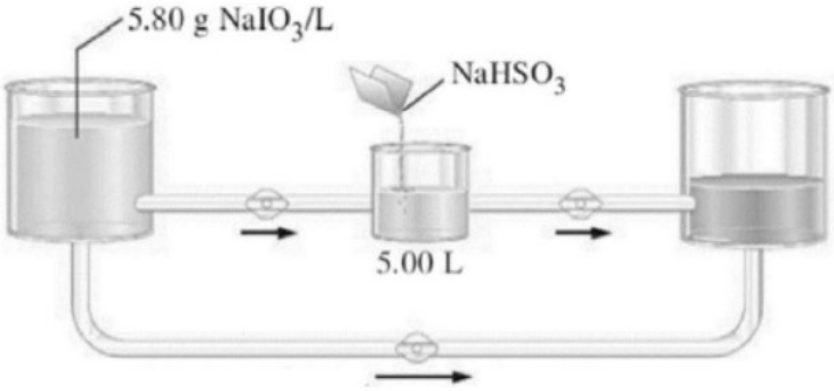
Question Stem for Question Nos. 49 and 50:

Chilie salt petre is a natural source of $NaNO_3$, it can also contains $NaIO_3$. The $NaIO_3$ can be used as a source of iodine. Iodine is produced from sodium iodate in two-step process occurring under acidic conditions:

$IO_3^-(aq) + HSO_3^-(aq) \rightarrow I^-(aq) + SO_4^{2-}(aq) + H^+(aq)$ (not balanced)

$IO_3^-(aq) + H^+(aq) + I^-(aq) \rightarrow I_2(s) + H_2O(l)$ (not balanced)

In the illustration, a 5.00 L sample of a $NaIO_3(aq)$ solution containing 5.80 g $NaIO_3$ / L is treated with the stoichiometric quantity of $NaHSO_3$ (no excess of either reactant). The, further quantity of the initial $NaIO_3(aq)$ is added to the reaction mixture to bring about the second reaction. [M.wts. $NaIO_3 = 198$ and $NaHSO_3 = 104\,g/mol$]



49. How many grams of $NaHSO_3$ are required in the first step? [Round off the answer to nearest integer]

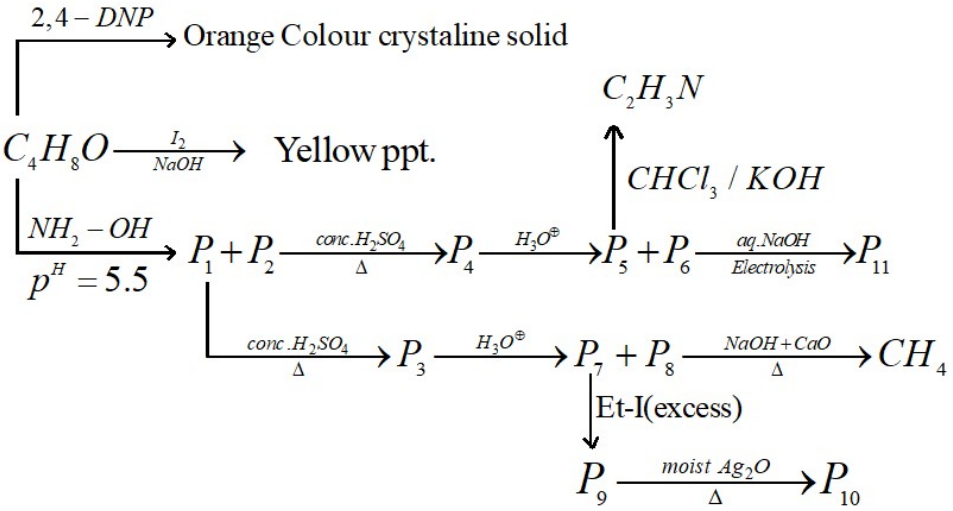
50. The additional volume (in ml) of the starting solution that must be added in the second step?

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SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer. For each question, choose the option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.
Negative Marks: –1 In all other cases.

Question Stem for Question Nos. 51 and 52:



51. Identify P_{11}

A) CH_4 B) C_2H_6 C) C_3H_8 D) C_4H_{10}

52. What is the correct about P_{10}

A) It can not give Bromine water test

B) It can not decolourize Bayer's reagent

C) It will give same product with HBr in presence or absence of peroxide

D) It can give red ppt. with Cu_2Cl_2 / NH_4OH

Question Stem for Question Nos. 53 and 54:

When a white crystalline compound (X) is heated with $K_2Cr_2O_7$ and concentrated H_2SO_4, a reddish brown gas (A) is evolved. On passing (A) into caustic soda solution, a yellow coloured solution of (B) is obtained. Neutralizing the solution of (B) with acetic acid and on subsequent addition of lead acetate a yellow precipitate (C) is obtained. When (X) is heated with NaOH solution, a colourless gas is evolved and on passing this gas into K_2HgI_4 solution a reddish brown precipitate (D) is formed.

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53. Which of the following is wrong ?

A) 'X' undergoes sublimation

B) The oxidation state of metal in the compound 'A' is +6

C) The aqueous solution of compound 'X' is basic in nature.

D) The compound 'C' is $PbCrO_4$.

54. Which of the following is correct?

A) Heating of 'X' with $NaOH$ is a redox reaction

B) The compound 'A' is CrO_2Cl_2

C) The gas liberated on heating 'X' with $NaOH$ turns blue litmus to red.

D) The compound 'X' is NH_4NO_3

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

55. A saturated solution in AgA ($K_{sp} = 3\times10^{-14}\ M^2$) and AgB ($K_{sp} = 1\times10^{-14}\ M^2$) has conductivity of $375\times10^{-10}\ S\,cm^{-1}$ and limiting molar conductivity of Ag^+ and A^- are $60\,S\,cm^2\,mol^{-1}$ and $80\,S\,cm^2\,mol^{-1}$ respectively then, the limiting molar conductivity of B^- is $\left[(30\times y)\,S\,cm^2\,mol^{-1}\right]$. The value of **'y'** is

56. The atomic masses of He and Ne are 4 and 20 a.m.u., respectively. The value of the de Broglie wavelength of He gas at -73^0C is "M" times that of the de Broglie wavelength of Ne at 727^0C. M is

57. How many of the given chemicals liberates dinitrogen on heating
$NH_4NO_2, NaN_3, (NH_4)_2Cr_2O_7, Pb(NO_3)_2, Ba(N_3)_2, (NH_4)_2SO_4, NH_4HPO_4$?

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